

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 8: Secondary Surveillance Radar, Mode S & ADS-B

Syllabus Source: CAE Oxford (pp. 297-340) | Pass Target: >= 90% AviationExam

1. SECONDARY SURVEILLANCE RADAR (SSR) PRINCIPLES

Unlike primary radar which depends on passive echoes, SSR relies on active cooperation between ground interrogator and aircraft transponder:

- Frequencies: Interrogation uplink = 1030 MHz (UHF). Transponder reply downlink = 1090 MHz (UHF). (Separate frequencies completely eliminate ground clutter and weather reflections!).
- Mode A (Identity): 4-digit octal code (0000 to 7777, 4,096 combinations). Special Emergency Squawks: 7700 (General Emergency / Mayday), 7600 (Radio Communications Failure), 7500 (Unlawful Interference / Hijacking!).
- Mode C (Pressure Altitude): Automatically encodes pressure altitude referenced to 1013.25 hPa in 100 ft increments (Gillham code). (Always reports standard pressure altitude, regardless of cockpit altimeter subscale setting!).

2. MODE S TRANSPONDER & ADS-B OUT

Mode S provides selective interrogation and high-capacity data link communication:

- Unique 24-Bit ICAO Address: Over 16 million unique aircraft addresses permanently assigned to the airframe registration. Eliminates identity confusion.
- Altitude Resolution: Reports altitude in 25 ft increments (compared to 100 ft in Mode C).
- ADS-B Out (Automatic Dependent Surveillance-Broadcast): The transponder automatically broadcasts aircraft GPS position, velocity, and intent over the 1090 MHz Extended Squitter once per second without requiring ground interrogation.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Transponder Squawk Codes & M**

Three Universal Emergency Codes: - 7500 -> Hijack / Unlawful Interference ('Seven-Five, man with a knife'). - 7600 -> Radio Communications Failure ('Seven-Six, radio needs a fix'). - 7700 -> General Emergency / Distress ('Seven-Seven, going to heaven'). Mode C / S Pressure Reference: - Transponder altitude encoder ALWAYS references standard 1013.25 hPa (Pressure Altitude) in 100 ft increments. - Ground ATC computers automatically correct this to local QNH.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What are the uplink (interrogation) and downlink (reply) frequencies used by Secondary Surveillance Radar?

- [A]: 108 MHz and 118 MHz
- [B]: 1030 MHz uplink and 1090 MHz downlink
- [C]: 960 MHz and 1215 MHz
- [D]: 5 GHz

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. By international standard, SSR interrogates on 1030 MHz and transponders reply on 1090 MHz.

Q2: When a transponder transmits MODE C altitude information to ATC, to what barometric pressure datum is

- [A]: Local aerodrome QNH
- [B]: STANDARD 1013.25 hPa (permanently, regardless of the subscale set on the altimeter)
- [C]: Aerodrome QFE
- [D]: Zero pressure

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Mode C reports uncorrected pressure altitude referenced strictly to standard 1013.25 hPa. Ground ATC radar computers convert this to indicated altitude using local QNH.

Q3: Which transponder squawk code MUST a pilot select immediately following a complete loss of two-way ra

- [A]: 7500
- [B]: 7600
- [C]: 7700
- [D]: 2000

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Emergency transponder codes: 7700 = Emergency; 7600 = Radio Communications Failure; 7500 = Hijack / Unlawful Interference.